

# Windows Longhorn Client Preview 2 Build 6.0.4074 AMD64

*A technical, research-driven assessment for Huey-VistaBox: provenance, NT 6.0 x64 behavior, virtualization, and PuTTY 0.83 compatibility*

## Executive summary

Windows Longhorn build **6.0.4074.idx02.040425-1535** (“**Longhorn Client Preview 2**”) is the **second and last widely distributed pre-reset Longhorn preview**, provided to attendees at **WinHEC 2004 (Seattle, May 4–7, 2004)** as a developer-facing preview for hardware/driver ecosystem preparation. <sup>1</sup> This build is historically “famous” because it is one of the clearest externally available snapshots of the pre-reset Longhorn vision—especially its composited desktop experiments and early “Aero” lineage—but *those headline UI characteristics are primarily true of the x86 build rather than AMD64*. <sup>2</sup>

For **AMD64 specifically**, BetaWiki’s evidence-based cataloging reports that **managed/.NET-adjacent UI features do not function** (explicitly including the **Sidebar**, **MIL effects**, and **Aurora**), which is a major constraint if your goal is “the most Vista-like Longhorn.” <sup>2</sup> The AMD64 4074 build can still be valuable as a “minimum viable NT 6.0 x64” environment, but you should treat it as a **kernel + Win32 compatibility testbed** more than an authentic “Aero demo box.”

For your stated must-have—**PuTTY 0.83**—the evidence is encouraging but not absolute. **PuTTY 0.83 (released 2025-02-08)** explicitly restored Windows XP compatibility after 0.82 temporarily broke it, and PuTTY’s own FAQ notes that 0.83 still runs on Windows XP even though 0.82 did not. <sup>3</sup> Since XP predates NT 6.0, this strongly suggests PuTTY 0.83 is conservative in its Win32 dependencies and should *in principle* run on Longhorn 4074—**especially if you use the standalone EXE instead of the MSI**, because Longhorn-era Windows Installer compatibility is repeatedly a pain point in vintage/pre-release Windows stacks. <sup>4</sup>

Operationally, your main technical risks in 4074 AMD64 for Huey-VistaBox are **network reliability and logon-critical services**: BetaWiki documents **DNS failures when DNS is obtained via DHCP**, and also warns that **Terminal Services and the User Profile Service are required for successful logon** (misconfiguring them can yield hangs/black screens or broken profiles). <sup>5</sup> These risks are manageable with a disciplined VM configuration and a reproducible “stabilization checklist.”

Finally, from a security and lab governance standpoint: you should treat pre-release Longhorn as **structurally unpatchable and unsafe for real data**. CISA’s general guidance on end-of-support Windows emphasizes that unsupported systems stop receiving security updates and become higher risk for compromise. <sup>6</sup> Your own HueyOS program framing also emphasizes **offline-first operation and careful security/data handling** in principle, which aligns with running Longhorn only in isolated test roles. <sup>7</sup>

## Introduction

Longhorn (the codename that ultimately became Windows Vista) is best understood as **two projects separated by a substantial reset**. The early (“pre-reset”) branch attempted to deliver a set of “pillars” that Microsoft publicly described in 2003 as:

- **Avalon** (new presentation/graphics subsystem),
- **WinFS** (new storage subsystem),
- **Indigo** (new communications/Web-services subsystem). <sup>8</sup>

At **WinHEC 2004**, Microsoft positioned Longhorn preview code as a practical on-ramp for OEMs and IHVs—especially to begin work on **native Longhorn device drivers** and to support forward-looking graphics/driver architectures. <sup>9</sup> Build 4074 is the canonical artifact of that moment: it is explicitly described as the WinHEC 2004 “Preview 2” build distributed at the conference. <sup>10</sup>

By **August 2004**, Microsoft publicly acknowledged schedule and scope changes (including shipping WinFS later) while targeting broad availability in 2006, reflecting the broader reset dynamics. <sup>11</sup> That reset matters for you because:

- **4074 is pre-reset**: it captures ambitious UI experiments but is also inherently incomplete and unstable.
- Later Vista-era builds (e.g., **5048, 5112, 5384.4**) are post-reset and typically more “Vista-like” *in day-to-day operation*, especially on AMD64. <sup>12</sup>

In your Huey-VistaBox context, HueyOS is described as an **offline-first OS/software layer** coordinating local AI, tools, and hardware—currently grounded in a real Linux-first baseline. <sup>7</sup> That architecture strongly suggests a Windows Longhorn guest should be treated as a **contained compatibility island** (e.g., legacy UI/driver research, “minimum x64 Windows” demonstrations, or controlled client tooling), not as a trusted operational node.

## Methods of research

This assessment uses a “primary-first, triangulated secondary” method.

First, your connected GitHub repository was reviewed to anchor assumptions about HueyOS and the intended lab posture (offline-first orientation, security/data-handling emphasis, and practical OS minimization). <sup>7</sup> A supporting internal note on OS “debloating” was also reviewed to align stabilization recommendations with your project’s general preference for lean systems. <sup>13</sup>

Second, build metadata and behavioral specifics were drawn from high-signal Longhorn archival compendia (notably **BetaWiki** and the **BetaArchive Wiki**) that catalogue build strings, compile dates, and known defects with unusually granular detail. <sup>2</sup>

Third, Microsoft primary sources were used to validate historical context (WinHEC 2004 positioning and Longhorn “pillars”) and to ground subsystem explanations (DWM and WOW64 behavior) in official technical descriptions. <sup>14</sup>

Fourth, PuTTY compatibility was assessed from PuTTY’s official release pages, changelog, and bug records (to avoid third-party repack uncertainty). 15

Throughout, where sources conflict or where the AMD64 build is known to diverge functionally from x86, the uncertainty is stated explicitly and operational guidance is framed as testable hypotheses rather than facts.

## Results

### Build identity and provenance

The build you selected is identified as:

- **Build string:** 6.0.4074.idx02.040425-1535
- **Lab:** idx02
- **Compile date:** 2004-04-25
- **Release name:** “Preview 2” / “Longhorn Client Preview 2”
- **Architectures reportedly compiled:** x86, AMD64, IA-64
- **Distribution:** given to WinHEC 2004 attendees in Seattle (May 4–7, 2004) and described as one of only two pre-reset Longhorn builds widely released by Microsoft (the other being 4051). 10

A concise metadata table (excluding activation/key material) is below.

| Property                         | Value (Build 4074 AMD64)                                                                              | Evidence |
|----------------------------------|-------------------------------------------------------------------------------------------------------|----------|
| Canonical build tag              | 6.0.4074.idx02.040425-1535                                                                            | 16       |
| Compile date                     | 2004-04-25                                                                                            | 16       |
| External release context         | WinHEC 2004 developer preview distribution                                                            | 17       |
| Time-limited evaluation behavior | Timebomb triggers after ~180 days from installation and results in forced logoff behavior             | 16       |
| High-level intent                | Early build for driver/hardware ecosystem prep; explicitly “very early” and performance not the focus | 18       |

### Pre-reset Longhorn’s UI/graphics thesis vs the AMD64 reality

Build 4074 is famous largely because the x86 build is reported to include:

- “complete” early MIL effects (e.g., transparency in Explorer surfaces when DWM is used),
- early Flip3D variant,
- an animated “stars” placeholder effect standing in for Aurora,
- Jade-themed DWM frame resources and other “Aero lineage” artifacts. 2

However, the most important AMD64-specific result for your selection is BetaWiki’s explicit caveat:

“features related to .NET Framework, such as the sidebar, MIL effects or Aurora, do not work on the x64 version.” <sup>2</sup>

That statement is load-bearing: it implies your AMD64 selection is not the “full 4074 experience” that most Longhorn visual retrospectives describe, because many of those visuals are bound to managed/Avalon-era components that are not functional in the x64 build.

A practical comparison (as a Huey-VistaBox operator) looks like this:

| Capability class                       | 4074 x86<br>(commonly documented)                        | 4074 AMD64 (your target)                                                                       | Operational meaning for Huey-VistaBox                                         |
|----------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Sidebar tiles / managed shell surfaces | Present and central to the “Longhorn feel” <sup>19</sup> | Reported non-functional <sup>16</sup>                                                          | AMD64 will feel “less Longhorn” out of the box; don’t plan UX around Sidebar  |
| MIL-driven translucent surfaces        | Reported as “complete MIL effects” <sup>16</sup>         | Reported non-functional <sup>16</sup>                                                          | Avoid making DWM/MIL part of the success criteria                             |
| Aurora (or placeholder equivalent)     | Present in x86 narratives <sup>2</sup>                   | Reported non-functional <sup>16</sup>                                                          | Treat Aurora as non-goal on AMD64                                             |
| Basic DWM enabling                     | Documented pathway exists <sup>16</sup>                  | BetaWiki suggests “basic DWM” can be enabled via direct calls/utility extraction <sup>16</sup> | DWM experimentation is possible but should be considered brittle and optional |

## NT 6.0 x64 mechanics that matter for “minimum viable Windows”

Even if you ignore the “Vista-like” UI layer, the AMD64 build is still a real NT-family 64-bit OS and inherits the fundamental constraints of x64 Windows that directly affect testbed compatibility.

- **WOW64 is the standard 32-bit compatibility subsystem** on 64-bit Windows, supporting 32-bit GUI/console/service apps while isolating 32/64-bit DLL worlds and preventing bitness mixing within a single process. <sup>20</sup>
- WOW64 uses **file system redirection** (e.g., redirects many 32-bit accesses targeting `%windir%\System32` to `%windir%\SysWow64`) to prevent 32-bit processes from loading 64-bit system binaries. <sup>21</sup>
- **64-bit Windows does not support 16-bit Windows applications**, and it does not support 32-bit kernel-mode drivers. <sup>22</sup>

For Huey-VistaBox, this implies:

1. A “modern-ish” 32-bit userland artifact like PuTTY 0.83 is *conceptually compatible* if WOW64 is present and stable.

2. Anything requiring legacy 16-bit installers (common in very old tooling) is a hard no.
3. Any device integration requiring unsigned/32-bit drivers is a hard no (and driver availability is already a core weakness of pre-release Windows).

### Known defects and reliability issues relevant to your use case

Build 4074's known issues are extensive, but the following are the ones most tightly coupled to a PuTTY-centric VM role:

- **DHCP-acquired DNS can break name resolution**, and the recommended mitigation is to **configure DNS manually**. <sup>5</sup>
- The OS **depends on Terminal Services and the User Profile Service for logon**; disabling or mis-setting these services can yield black-screen hangs or broken profiles/Explorer initialization failures. <sup>23</sup>
- There are reports of **severe hangs shortly after logon in some multi-core configurations**, suggesting you should treat CPU-core count as a stability lever (start low, scale up only if stable). <sup>23</sup>
- **Internet Explorer's built-in download manager is reported broken**, with downloads not working at all in some cases—meaning you should not rely on in-guest web retrieval for PuTTY or drivers. <sup>16</sup>
- Setup/WinPE has input pitfalls: **PS/2 mouse driver is broken in the Preinstallation Environment**, which directly affects VM installs because many virtual pointing devices present as PS/2 by default. <sup>16</sup>
- The build contains an evaluation **timebomb** (~180 days from installation) that forces logoff/blocks use after expiration. <sup>16</sup>

### Virtualization behavior and recommended VM configuration

BetaWiki's VM guide provides two kinds of virtualization guidance: (a) hypervisor-wide caveats for vintage Windows, and (b) configuration templates for specific Windows eras/build families.

Key caveats to take seriously:

- Older Windows builds may run better on **older VMware compatibility modes**, and later host/hypervisor behaviors can break DWM or cause instability. <sup>24</sup>
- VMware Easy Install can mis-detect AMD64 Longhorn builds as x86 XP and attempt to inject wrong drivers, producing setup errors. <sup>24</sup>
- DWM can be fragile in virtualization, including black screens or crashes on some host systems. <sup>24</sup>

A pragmatic baseline for **Milestone 7 pre-reset Longhorn x64** (the family containing 4074) in VMware is:

- 1 VM, **~1.5 GB RAM, 1 CPU / 2 cores**, IDE disk (~25 GB), USB 2.0, video memory ~128 MB, 3D acceleration enabled, OS type set to "Windows XP Professional x64 Edition," hardware compatibility Workstation 6.5–7.x. <sup>25</sup>
- For sound and accelerated graphics features, the guidance emphasizes **VMware 6.5.2 driver packages extracted from VMware guest additions**, rather than modern VMware Tools. <sup>26</sup>

Notably, longhorn.ms (community analysis) documents a useful operational distinction: **full VMware Tools may not be compatible, but extracted driver packages often are**, and it provides VMX-level tweaks to

avoid host-driven time sync and to work around CPU presentation issues on some modern Ryzen systems.

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## PuTTY 0.83 compatibility assessment for Longhorn 4074 AMD64

From PuTTY's official release materials:

- **PuTTY 0.83** is the latest release as of its release page, dated **2025-02-08**, and offers both MSI installers and standalone executables (including a direct `putty.exe` download). 28
- PuTTY's official changelog states: **"Windows XP support was accidentally broken, and now reinstated."** 29
- PuTTY's wishlist bug record details how 0.82 binaries failed on XP due to a manifest-related issue and that this was fixed in 0.83. 30
- The PuTTY FAQ clarifies that the project targets currently supported Windows, but older versions may still work; explicitly: **"PuTTY 0.83 still runs on Windows XP – but the previous release 0.82 didn't."** 31

**Inference (clearly labeled):** If PuTTY 0.83 is engineered to remain compatible with Windows XP-era constraints, then it is likely to operate on Longhorn 4074's Win32 surface area as well—particularly as a standalone EXE—because Longhorn 4074 is still a classic Win32 desktop and (as an NT-family successor-era build) should not be *less* capable than XP in baseline user-mode APIs. This inference is supported indirectly by PuTTY's XP-compatibility posture and by Microsoft's general compatibility framing of WOW64 on x64 Windows. 32

The practical risk is less "does PuTTY run?" and more "does networking behave?" because DNS/DHCP issues are specifically documented in 4074. 23

## Discussion

### Is 4074 AMD64 the best "polished / Vista-like Longhorn"?

As an expert-to-expert judgment grounded in the cited evidence: **no—at least not on AMD64.**

Build **4074 x86** is the variant most associated with "complete" pre-reset MIL/DWM visual experiments. 2  
But the **AMD64** build is explicitly documented as missing/breaking core managed UI features (Sidebar/MIL/Aurora), which are central to that "Vista-like Longhorn" identity. 16

If your actual goal is **"minimum 64-bit Windows OS that can run PuTTY"**, 4074 AMD64 can still be defensible because:

- It is a real NT 6.0 x64 environment (useful architecturally as a boundary case),
- It has a strong historical pedigree and well-documented quirks,
- You can design Huey-VistaBox around **PuTTY + a small controlled toolset**, not around Longhorn eye-candy.

If your goal shifts back toward **“Vista-like” on AMD64**, later post-reset builds are structurally better candidates:

- **5048** (WinHEC 2005 “Longhorn Developer Preview”) is post-reset and noted as more XP/2003-like, but it’s still an evaluation-style build and can be installed currently; it also has explicit AMD64 compile notes. <sup>33</sup>
- **5112** (Vista Beta 1) is the first post-reset build to become widely available; DWM can be enabled with known workarounds, which makes it a more plausible “Vista-like” staging environment. <sup>34</sup>
- **5384.4** (Vista Beta 2, May 2006) is much closer to the eventual Vista UX and generally has a more mature driver/software posture than pre-reset Longhorn. <sup>35</sup>

### Build comparison table requested

The table below compares **4074 AMD64 vs 4074 x86 vs later AMD64 builds** across the attributes you asked for. When a cell is an inference rather than explicitly stated, it is marked **(inferred)**.

| Attribute                       | 4074 AMD64                                                | 4074 x86                                                             | 5048 AMD64                                                          | 5112 AMD64                                                   | 5384.4 AMD64                                                  |
|---------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------|
| Development era                 | Pre-reset Longhorn (WinHEC 2004 preview) <sup>16</sup>    | Same <sup>16</sup>                                                   | Post-reset (WinHEC 2005 dev preview) <sup>33</sup>                  | Post-reset Vista Beta 1 <sup>34</sup>                        | Post-reset Vista Beta 2 <sup>35</sup>                         |
| “Vista-like” visuals out of box | Limited by missing managed UI features <sup>16</sup>      | Strongest pre-reset “Aero lineage” build commonly cited <sup>2</sup> | More XP/2003-like (post-reset early) <sup>33</sup>                  | More Vista-like; DWM possible with workarounds <sup>34</sup> | Closest to Vista UX among these <sup>35</sup>                 |
| Sidebar / MIL / Aurora          | Reported non-functional <sup>16</sup>                     | Present/featured <sup>2</sup>                                        | Not emphasized; post-reset refactoring era (inferred) <sup>33</sup> | Present via DWM enabling path (inferred) <sup>34</sup>       | Present per Beta 2 Vista feature set (inferred) <sup>35</sup> |
| Stability posture (relative)    | Known serious quirks; pre-release fragility <sup>23</sup> | Similar class of quirks <sup>23</sup>                                | Early post-reset; still rough but reset-driven <sup>33</sup>        | Beta-class; still time-limited but more mature <sup>34</sup> | Beta 2; generally more mature than Beta 1 <sup>35</sup>       |
| Time limitation                 | ~180-day timebomb from install <sup>16</sup>              | Same <sup>16</sup>                                                   | ~180-day timebomb from install <sup>33</sup>                        | Fixed expiration date (timebomb date) <sup>34</sup>          | Fixed expiration date (timebomb date) <sup>35</sup>           |

| Attribute                        | 4074 AMD64                                                                                         | 4074 x86                                                             | 5048 AMD64                                                                | 5112 AMD64                                             | 5384.4 AMD64                                                         |
|----------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------|
| Virtualization friendliness      | Usable with careful config; DWM fragile; manual driver packs often required <sup>36</sup>          | Often used for “Aero in VM” experiments; still fragile <sup>37</sup> | (inferred) More VM-friendly than pre-reset, but still early <sup>33</sup> | DWM often needs patched components in VM <sup>34</sup> | Generally VM-friendly compared to pre-reset (inferred) <sup>35</sup> |
| Software compatibility (general) | NT6 x64 constraints apply; many modern installers may fail; best with portable tools <sup>38</sup> | Similar, but x86 app ecosystem often easier                          | Better than pre-reset for mainstream Win32 (inferred) <sup>33</sup>       | Better than pre-reset (inferred) <sup>34</sup>         | Best among these four for broad software (inferred) <sup>35</sup>    |

### Virtualization guidance for Huey-VistaBox

Given Huey-VistaBox’s likely role (a demonstrative/compatibility VM), prioritize: reproducibility, deterministic networking, and “portable apps first.”

- **Prefer VMware over VirtualBox if you care about any DWM experiments**, but expect fragility regardless; BetaWiki explicitly warns that DWM may render artifacts, black screens, or crashes depending on host hardware and VM settings. <sup>24</sup>
- **Use older compatibility modes** (Workstation 6.5–7.x hardware compatibility in VMware terms) where possible; multiple sources and community practice indicate setup/driver behavior can fail if you present too modern a virtual hardware profile. <sup>39</sup>
- **Treat extracted driver packages as the first-line path**, not modern “Tools,” consistent with longhorn.ms’s note that Tools may be incompatible while drivers remain usable. <sup>40</sup>

### Security and forensics: why isolation is not optional

Even though Longhorn 4074 is a research target, it should be treated as **high-risk software**:

- It is not supported and cannot receive modern security updates; CISA’s end-of-support guidance for Windows makes the general point that unsupported systems no longer receive updates/hotfixes and become more vulnerable. <sup>6</sup>
- HueyOS’s own framing emphasizes that any system capable of processing/transmitting data carries risk and that risk increases when network access is enabled—an especially relevant reminder when running a pre-release Windows VM. <sup>7</sup>

So: keep Huey-VistaBox behind NAT or host-only networking, avoid exposing services, and never store secrets in the guest.

### Longhorn timeline around 4008–5384



```

flowchart LR
    A[4008<br/>M4<br/>2003-02-19] --> B[4051<br/>PDC 2003 Preview<br/>2003-10-01]
    B --> C[4074<br/>WinHEC 2004 Preview 2<br/>2004-04-25]
    C --> D[Reset dynamics<br/>2004 public re-scope<br/>WinFS deferred]
    D --> E[5048<br/>WinHEC 2005 Dev Preview<br/>2005-04-01]
    E --> F[5112<br/>Vista Beta 1<br/>2005-07-20]
    F --> G[5384.4<br/>Vista Beta 2<br/>2006-05-18]

```

Key anchor points and dates are taken from BetaWiki build pages and Microsoft's 2004 announcement about deferring WinFS beyond Longhorn. <sup>41</sup>

## Conclusion

Windows Longhorn **6.0.4074 AMD64** is a historically significant WinHEC 2004 preview build with a well-mapped defect surface, making it a legitimate candidate for a **research VM** and a “minimum NT 6.0 x64” demonstration OS. <sup>42</sup>

However, it is **not** the best choice if your primary criterion is “most polished / Vista-like Longhorn,” because the AMD64 build is explicitly documented as lacking functional Sidebar/MIL/Aurora features that define the famous 4074 experience. <sup>2</sup>

For your PuTTY requirement, PuTTY **0.83** is a strategically good target because it intentionally maintains XP compatibility and offers standalone executables, which is exactly the posture you want when you cannot trust MSI/installer stacks in a pre-release OS. <sup>3</sup> The primary engineering work in Huey-VistaBox will likely be **network stabilization** (manual DNS, conservative VM NIC choices) and **service discipline** (do not disable logon-critical services).

## Appendices

### Appendix: recommended VM configuration profiles

#### VMware profile (research baseline for 4074 AMD64)

Start from BetaWiki's Milestone 7 pre-reset Longhorn x64 guidance as a baseline (then tune for stability):

- Memory: ~1.5 GB
- CPU: start with 1 core (then test 2 cores only if stable, because multi-core hangs are reported) <sup>43</sup>
- Disk: IDE (≈25 GB) <sup>44</sup>
- USB: enabled (helps around WinPE input edge cases) <sup>45</sup>
- Avoid Easy Install autoinjection behaviors for AMD64 Longhorn builds <sup>24</sup>
- Drivers: prefer extracted VMware 6.5/6.5.2-era driver packages rather than modern Tools <sup>27</sup>

#### VMX tweaks (time discipline + CPU presentation)

Community analysis suggests disabling VMware time sync (useful for deterministic testing) and overriding

CPUID presentation if early builds fail to boot on modern hosts. <sup>40</sup>  
(These are configuration concepts; apply only if you see the specific failure modes.)

### **VirtualBox profile (PuTTY-first, visuals not a goal)**

Given that Longhorn DWM/Aero paths can be unreliable on VirtualBox for early builds, treat VirtualBox as “good enough” if the objective is simply PuTTY + networking, not composited UI experiments. The moment DWM becomes a requirement, VMware is generally the more workable path per BetaWiki’s virtualization notes. <sup>24</sup>

## **Appendix: reproducible install + stabilization checklist for Huey-VistaBox (4074 AMD64)**

### **1. Create VM + snapshot plan**

2. Create VM with conservative virtual hardware (IDE disk, minimal devices).
3. Disable “easy install” / unattended behaviors that may inject mismatched drivers. <sup>24</sup>
4. Define snapshot points: *post-install boot*, *post-network stabilization*, *post-driver install*, *post-PuTTY validation*.

### **5. Install OS (WinPE ergonomics)**

6. Expect WinPE input friction: PS/2 mouse in WinPE is documented broken; plan to navigate setup with keyboard or a USB-pointing configuration. <sup>16</sup>

### **7. First boot sanity checks (do not optimize yet)**

8. Confirm you can log on repeatedly without hangs.
9. **Do not disable services** during early stabilization.

### **10. Logon-critical service checks**

11. Ensure **Terminal Services** and **User Profile Service** remain enabled/automatic because the build is documented as relying on them for correct logon/session initialization. <sup>23</sup>

### **12. Networking: fix DNS before blaming PuTTY**

13. If you can ping by IP but not resolve hostnames, set a **manual DNS server** rather than DHCP-provided DNS (explicitly documented as a failure mode). <sup>23</sup>
14. Prefer NAT or host-only networking; avoid bridged exposure unless you have a controlled lab VLAN.

### **15. Graphics/sound drivers (optional for PuTTY; required only for UI experiments)**

16. If you need improved resolution or sound, use extracted VMware driver packages rather than assuming modern Tools are compatible. <sup>27</sup>
17. If you attempt DWM experimentation, treat it as best-effort; DWM can artifact or crash depending on host hardware. <sup>24</sup>
18. **Install PuTTY 0.83 (portable-first)**
19. Avoid MSI. Use the official **standalone executable** download route from PuTTY's site (x64 first, fall back to x86 if needed). <sup>28</sup>
20. Rationale: PuTTY 0.83 is deliberately conservative enough to still run on XP, implying it avoids some newer dependency traps. <sup>46</sup>
21. **PuTTY validation tests (define "works")**
22. DNS test: connect by hostname and by IP; record differences (hostname failures usually imply DNS). <sup>23</sup>
23. SSH handshake test: verify key exchange completion and interactive shell.
24. Cipher/key stress: optional, but note 0.83 adds modern KEX (ML-KEM); older CPUs may be slow but should function. <sup>29</sup>
25. **Security hardening for lab use**
26. Treat the VM as untrusted: no secrets, no personal accounts, no direct exposure. <sup>6</sup>
27. Align with HueyOS posture: prefer offline/isolated operation and minimize unnecessary services, but **never** at the cost of disabling logon-critical services in this specific build. <sup>47</sup>

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<sup>1</sup> <sup>2</sup> <sup>5</sup> <sup>10</sup> <sup>16</sup> <sup>17</sup> <sup>23</sup> <sup>42</sup> <sup>43</sup> <sup>45</sup> <sup>47</sup> [https://betawiki.net/wiki/Windows\\_Longhorn\\_build\\_4074](https://betawiki.net/wiki/Windows_Longhorn_build_4074)  
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